

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing

Year 1, Semester 1- 2024

Home Automation System - Pet Feeder

Proposal Document

P10-10

IT Number	Name
IT24101357	BANDARA N.H.M.W.W.D.S.
IT24101620	KALUARACHCHI K.A.T.H.
IT24102021	GUNASINGHE B.A.C.M.
IT24102947	RAMIN V.P.
IT24100542	ABDEEN M.A.L.
IT24103279	DUSHMANTHI W.D.H.

Contents

1. <i>Background</i>	3
2. <i>Problem and Motivation</i>	3
3. <i>Aim and Objectives</i>	4
4. <i>System Diagram</i>	5
5. <i>Methodology</i>	6
6. <i>Evaluation Method</i>	10
7. <i>References</i>	11



1. Background

The emergence of IOT (internet of things) has carved out a new sphere of unique function in the world of home automation. Among the leading fields, pet care is a standout one. Automatic pet feeders can help in this process. We envision this undertaking as creating an IoT comprised automatic pet feeder system of Arduino board. For most times, the gadget will be on and that way pets are in no danger of not eating on time whatsoever, even during trips and other engagements of the owners. Offering both a secure and comfortable solution, this device will use sensors, actuators, and communication modules which connect the pet owners virtually and thus the owners can also give the right amount of food according to their pet's schedule. It is done with the help of a remote and it is not only monitored but also controlled.

2. Problem and Motivation

Problem:

Pet owners often find it difficult to keeping their pets' feeding schedules due to their hectic schedules. Inconsistent diet is responsible for problems such as overweight, undernourished, and anxious pets. The characteristics of flexibility, control, and monitoring are necessary to adapt to the modern pet owner's ever-changing schedules.

Motivation:

An IOT-based smart feeder for pets will be a great advantage for pet owners since it will make sure that pets receive food regularly and in the right amounts, thus, this device reduces the possibility of any health problems. Pet owners can rest calm now that their pet is not only being fed on time by the automatic feeder, but they can also monitor and control their feeding schedule from any place in the world. This will in turn lead to much greater savings of time and money for the pet s e d instance of pet owner, which in turn is a great contributor for the pet to live a healthy and happy life. This device has such advantages as remote access, health monitoring, error alerts, and historical data access.

3. Aim and Objectives

Aim:

To build an automatic feeder system that offers remote control and monitoring via an online platform using IOT and guaranteeing that pets will feed on time.

Objective:

1. Hardware Design and Assembly

You will need a Arduino Uno, load cell with HX711 module, servo motor, time-of-flight sensor, ESP32 Wi-Fi Module and few other components that can easily be found in the market. Link the elements on your system design. (Figure 4.1)

2. Develop Microcontroller code

Write a code to dispense food with the servo motor. Set up the weight measurement via Load cell and HX711 Module. Use RTC module to integrate feeding schedules and the ESP32 module for remote control and monitoring.

3. Create user interface

Develop the Blynk app interface that allows the user to set up the feeding schedules, portion sizes and to monitor the system remotely. Add the options to notify for feeding times, low food levels and other errors.

4. Integrate and Test System

Integrate the hardware and software aspects of the systems. Conduct extensive testing that includes functionality and reliability. Debug and resolve any issues.

5. Deploy and user training

Deploy the system in a real-world scenario and train/educate the user to use it appropriately.

4. System Diagram

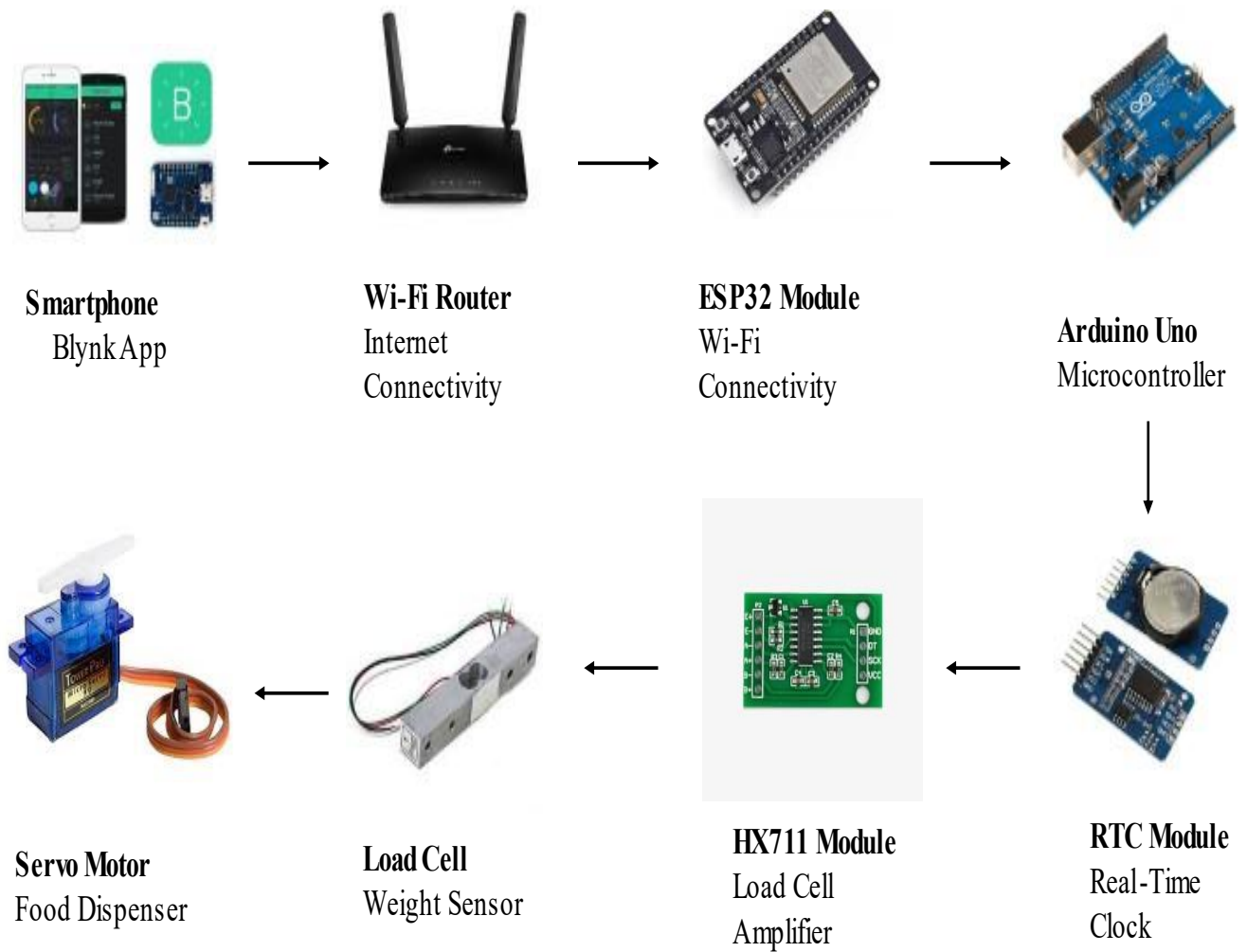


Figure 4.1: System diagram

5. Methodology

- **Hardware Assembly** - Connect and set up all components in accordance with the system diagram (Figure 4.1).
- **Software Development** – Write and test microcontroller code for Arduino and develop a connection between the Blynk app.
- **Integration** – Combine all the hardware and software components together so that all of these are in sync.
- **Testing and Debugging** – Conduct tests order to identify and fix those issues.

Hardware components:

- **Arduino uno:** The Arduino Uno's convenient programming environment and vast community support make it a popular microcontroller. Weight sensor, servo motor, RTC module and communication modules are all interconnected with the GPIO (General Purpose Input/Output) pins that on the Arduino Uno board. The Arduino Uno can guarantee that all connections are established without the need for any extra hardware or software.
- **Servo motor:** To effect precise control over mechanical motions, employ a servo motor like those used in modern CD players. In organizing corresponding exact control responsibilities on both sides of a parallelogram, this mechanism, and various others currently under foreign development operate accordingly. The last stage of the food distribution component can be easily programmed to rotate at any angle using this servo motor since it directly drives that component. Our breeders have taken advantage of their experience in this respect after many years of practice. The accuracy just described helps ensure that every unit of food is dispensed the same every time.
- **Weight sensor and HX711 module:** The HX711 module and weight sensor (load cell) are used by this machine so that food can be accurately dispensed. Doing so makes it less likely to overfeed or underfeed, because the right amount of food is always given out and there are no mistakes because of human error. The HX711 module amplifies the minute analog signal from a load cell and turns it into digital data that an Arduino can read.
- **Real-time clock module:** The RTC module doesn't keep track like a wall clock, but rather just a little more than its own moment of birth until it's unplugged from power. This is necessary for ensuring an even food supply. With the time from this module, the Arduino can actuate dispensing mechanism automatically at prescribed times.

- **ESP32 Wi-Fi module:** Connectivity Thx to module, the pet feeder can be managed over the Internet and accessed remotely from anywhere. With a computer or smartphone on hand to monitor feeding status, It's even possible for you from England. To manually release food and adjust the time and feeding schedules on the fly is also feasible using any network location.
- **Jumper wires:** During the development stage, these are essential for connecting the Arduino board with various components on your breadboard. They allow quick troubleshooting and changes of circuit configuration.
- **Breadboard:** Without soldering, the breadboard allows you to create a working prototype for this circuit. It makes simple to test or alter connections between Arduino, sensors., and parts like relays etc. This is particularly helpful in early development stages where it's possible that design may undergo large changes later without soldering the breadboard is used to build a working prototype for the circuit. It makes it easy to see and change how Arduino, sensors, or radios are connected. This is particularly useful during those early stages of development when you feel the design could well alter.

Software Tools and Technologies:

- **Arduino IDE** – The Arduino Development Environment (IDE) is a great platform for developing, compiling, uploading and debugging the code onto your Arduino microcontroller. Friendly to both beginners and old hands because of its friendly user interface, huge library collection and numerous illustrations. There are lots of people using the Arduino IDE, so there's also a sizable fan base really helpful if you want to solve common problems using internet resources. It has been in use for many years and countless people have written libraries for it that you can now get online right from the start.
- **HX711 Library** - The HX711 library makes reading data from the HX711 Load Cell Amplifier easier. It handles the data transfer between your Arduino and the HX711 module, translating the analog impulses of the load cell into digital values that can be read off. The project's implementation of a load cell is greatly facilitated using the library's simple utilities for calibrating and measuring weights for instance.
- **BLYNK** – It is a web page that allows the developer to create an application in mobile which will help him/her to control hardware remotely. It provides a drag-and-drop interface and pre-built widgets for various controls & displays to build the program. With the real-time connection between your phone and hardware you can do amazing things like update a display immediately or use buttons to give commands. Important as remote monitoring and hope for the pet feeder.
- **Servo Library** –The Arduino IDE's Servo Library Functions for controlling the position of a servo motor. It generates the oases of PWM (Pulse Width Modulation) that required to move the servo at specified locations. The library simplifies the servo motor integration with the Arduino to ensure an exact and smooth operation of this food dispenser.
- **RTC Library** – Arduino would use the Real-Time Clock library to perform some tasks such as reading the current date and time. This way of working ensures that time-based operations on the Arduino operate perfectly accurately. Even when the computer is shut down, precise feeding schedules are maintained by the library's ability to keep current time.
- **ESP32 WI-FI Library** – The ESP32 Wi-Fi library, which allows for the tools necessary to link an ESP32 module to a wireless network and permit remote control of a device (e.g., Arduino), data transmission in and out of Internet services are developed as well

with this system. It contains functions for operating HTTP requests dashboards for creating a new Wi-Fi connection and so on cloud service access routines give ways to interact online via a cloud host live. That means getting online from right wherever elsewhere you have both its data cloud availability and an open Internet connection. And all of this is made much easier to incorporate in deployments with Wi-Fi by the library.

Steps:

1. Set up the scheduled feeding and manual feeding via the Blynk app on the web interface or smartphone and transmits to Wi – Fi Router, which in turn sends them over the internet.
2. ESP32 Module receives the commands sends from the web interface or smartphone and communicate these with the Arduino Uno.
3. RTS check the feeding schedule and starts the servo motor to ensure the feeder is on time to release the food out. Measure food *dispensed* weight by the load cell and the load cell sends signals to the HX711 module.
4. The Arduino Uno reads the signal from the HX711 module in order to dispense the correct amount of food, If the amount is less or more than the assigned amount, the Arduino will adjust the servo motor's operation and sends back any relevant data back to the ESP32 module.
5. Then the ESP32 module send these data back to the Blynk app on the smartphone or the web interface via the Wi-Fi router

6.Evaluation Method

The system will be evaluated on the following criteria:

- **Usability Testing** : Monitoring users completing tasks , report difficulties. Focus on usability and errors.
- **A/B Testing** : A/B testing compares versions of interface to find preferred design, testing user preferences.
- **Surveys and Feedback** : Gather user feedback through polls and feature requests for app improvements.
- **Accessibility Testing** : Create designs accommodating disabilities, prioritize accessibility features, including color contrast.
- **Performance Testing** :Testing Application speed and reliability
- **Field Testing**: Observe, analyze users natural and random actions.

7. References

- [1] Researchgate.net. [Online]. Available: [https://www.researchgate.net/publication/349798792 Smart Pet Feeder](https://www.researchgate.net/publication/349798792_Smart_Pet_Feeder) [Accessed: 10-Aug-2024].
- [2] Iop.org. [Online]. Available: <https://iopscience.iop.org/article/10.1088/1757-899X/670/1/012069/meta#skip-to-content-link-target>. [Accessed: 10-Aug-2024].
- [3] P. N. Vrishanka, P. Prabhakar, D. Shet, and K. Rupali, “Automated Pet Feeder using IoT,” in 2021 IEEE International Conference on Mobile Networks and Wireless Communications (ICMNWC), 2021, pp. 1–5.
- [4] C. Pfister, Getting started with the internet of things: Connecting sensors and microcontrollers to the cloud. Sebastopol, CA: Maker Media, 2011.
- [5] “Automatic Pet Feeder using Arduino,” Circuitdigest.com. [Online]. Available: <https://circuitdigest.com/microcontroller-projects/automatic-pet-feeder-using-arduino>. [Accessed: 10-Aug-2024].
- [6] Techpins, “Automatic Arduino Pet Feeder,” Instructables, 23-Feb-2017. [Online]. Available: <https://www.instructables.com/Automatic-Arduino-Pet-Feeder/>. [Accessed: 10-Aug-2024].
- [7] “How to build a simple Arduino-based automatic pet feeder,” circuito.io blog, 10-Aug-2018. [Online]. Available: <https://www.circuito.io/blog/automatic-pet-feeder/>. [Accessed: 10-Aug-2024].